Mean $= \frac{3\times2+5\times4+3\times6+2\times7}{3+5+3+2} = \frac{6+20+18+14}{13} = 4.46$

Mode = 4, which has the highest frequency

Median, since there are 13 results, the median will be the 7th result.

In order there are 3 twos and 5 fours, the seventh result (in order) must be 4.

Exercise 7F

1. The table below shows the number of goals scored in a number of football matches.

Number of goals	0	1	2	3
Number of Matches (Frequency)	2	6	7	5

Calculate the mean number of goals scored in all the matches.
Give your answer correct to the nearest whole number.

2. The price of the same packet of biscuits in 20 different shops is shown in the table.

Price of biscuits	50c	51c	52c	53c	54c
Number of shops (Frequency)	2	3	6	5	4

(i) What is the mean price of the packet of biscuits?
Give your answer correct to 1 place of decimals.
(ii) What was the median price for the packet of biscuits?
(iii) What was the modal price for the packet of biscuits
(iv) How many shops charged 53c or more per packet?

3. The weekly maths test ("out of 10") produced the following frequency table.

Number of correct answers	1	2	3	4	5	6	7	8	9	10
Number of students (Frequency)	1	1	4	3	4	5	6	3	2	1

(i) How many students did the test?
(ii) How many students got 6 (out of 10) correct?
(iii) How many students got 6 or more (out of 10) correct?
(iv) What was the modal mark?
(v) What was the mean mark?

4. Find the mean, mode and median for the following frequency distribution.

Number	14	15	16	17	18
Frequency	2	5	6	4	1

Give the value of the mean correct to 1 place of decimals.

5. In a Biology class soil samples were taken for examination:

Weight (grams)	64	70	72	75	76
Frequency	2	5	4	3	1

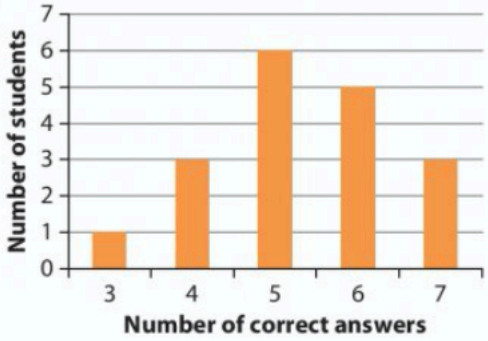
(i) Find the mean weight of the samples in the frequency table above.
Give your answer correct to 2 significant figures.
(ii) What was the modal weight?
(iii) How many samples weighed less than 70 g
(iv) What was the median weight of sample?

6. The results of a maths test are shown in a bar chart. Find:

(i) the modal number of correct answers.

(ii) the median number of correct answers.

By converting the chart to a frequency table or otherwise, calculate the mean number of correct scores.



Grouped Frequency Tables

When dealing with a large amount of data, results that are close together can be arranged in groups or classes.

The grouped frequency table below, shows the marks (out of 25) achieved by 24 students in a test.

Marks	1–5	6–10	11–15	16–20	21–25
Frequency	3	5	8	4	4

It is not possible to find an exact mean of a grouped frequency distribution because we do not have the raw data. We can find an **estimate** of the mean by taking **mid-interval values** of each class. The mid-interval values are found by adding 1 and 5 and dividing by 2, i.e.

$3 = \left(\frac{1+5}{2}\right), 8 = \left(\frac{6+10}{2}\right), 13 = \left(\frac{11+15}{2}\right)$ etc.

Mid-interval Marks	3	8	13	18	23
Frequency	3	5	8	4	4

The estimated mean is

$$\frac{3 \times 3 + 5 \times 8 + 8 \times 13 + 4 \times 18 + 4 \times 23}{3+5+8+4+4} = \frac{9 + 40 + 104 + 72 + 92}{24} = 13.2 \text{ marks}$$

Exercise 7F (continued)

7. In one hour 20 buses arriving at a depot were late. The number of minutes the buses were late are shown in the grouped frequency table below. (0–2 means times up to 2 minutes etc)

No. of minutes late	0–2	2–4	4–6	6–10	10–20
No. of buses	2	3	8	4	3

(i) Use mid-interval values to calculate the mean number of minutes the buses were late.

(ii) What was the "modal class" for this data?

(iii) In which interval would you find the median value?

(iv) What was the maximum number of buses that were more than 5 minutes late?

8. The following table shows the weights of parcels taken at a post office on a particular day.

Weight in kg	0–2	2–4	4–6	6–8	8–10
No. of parcels	6	12	20	48	14

(Where, 2–4, includes 2 kg but not 4 kg etc.):

(i) Use mid-interval values to calculate an estimate of the mean weight of the parcels, correct to the nearest kilogram.

(ii) What was the modal class of this data?

(iii) In which class does the median value lie?

(iv) What was the minimum number of parcels weighing 7 kg or less?